Let P and Q stand for statements. Which, if any, of the following statements follows from of the others?

(a) $P \to Q$
(b) $Q \to P$
(c) $\neg P \to \neg Q$
(d) $\neg Q \to \neg P$

We want to show that (d) follows from (a).

Suppose that (a) is true. That is, $P \to Q$ is true.

Now suppose that $\neg Q$ is true and that P is true.

Since $P \to Q$ is true and P is true it must be that Q is true which violates our supposition that $\neg Q$ is true.

So it must be that P is false so $\neg P$ is true and therefore $\neg Q \to \neg P$ is true.

(c) follows (b) is proven in an identical manner and vice-versa.